

25.2 Cosmology

Question Paper

Course	CIEA Level Physics
Section	25. Astronomy & Cosmology
Topic	25.2 Cosmology
Difficulty	Hard

Time allowed: 50

Score: /35

Percentage: /100

Question 1a

Fig. 1.1a shows the variation of recession velocity v for a number of galaxies with their approximate distance d from Earth.

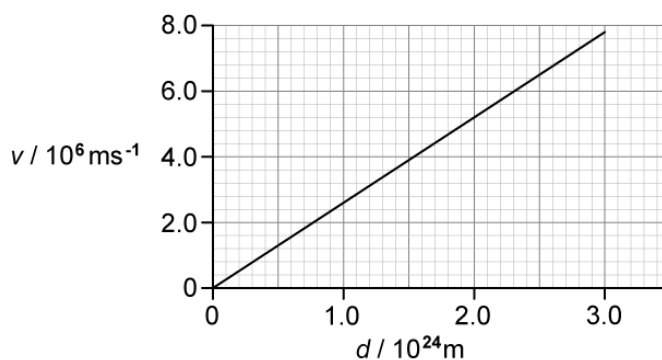


Fig. 1.1a

Radiation from a star in a distant galaxy is analysed.

Fig. 1.1b shows the spectral lines of the radiation from the star at different wavelengths compared to the same spectral lines produced by a source in the laboratory.

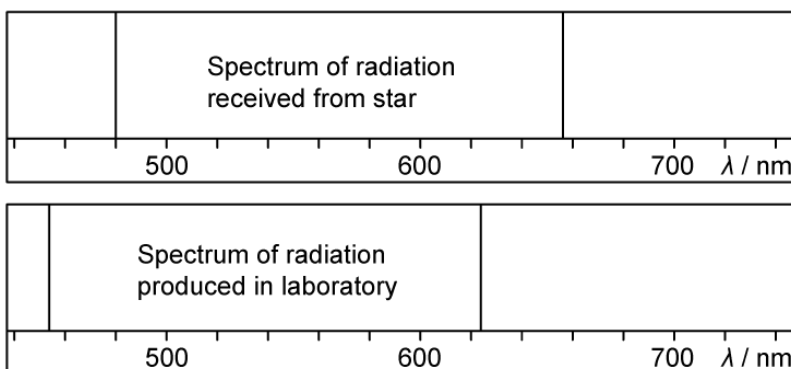


Fig 1.1b

Using the information in Fig. 1.1a and 1.1b, determine the distance of this galaxy from Earth.

[5 marks]

Question 1b

Astronomers use the cosmic scale factor R to describe the scaling up of physical distances in the Universe.

As the universe expands, the wavelength of the detected radiation at a time t is larger than it was at a time t_0 in the past by a factor of

$$\frac{R}{R_0} = 1 + z$$

This expression represents the expansion rate of the universe.

Where R represents the separation of two galaxies at time t , R_0 represents the value of R at the time t_0 the radiation was emitted, and z represents the redshift, which is given by

$$z \approx \frac{\Delta\lambda}{\lambda} \approx \frac{v}{c}$$

(i)

Using the scale factor, or otherwise, determine, in years, the approximate age of the universe at the instant when the detected light from the distant galaxy in **(a)** was emitted.

[4]

(ii)

State the assumption that you made in your answer to **(b)(i)**.

[1]

[5 marks]

Question 1c

Historically, there has been considerable debate over the true value of the Hubble constant.

Fig. 1.2 shows Hubble's original plot of velocity against distance for 46 galaxies.

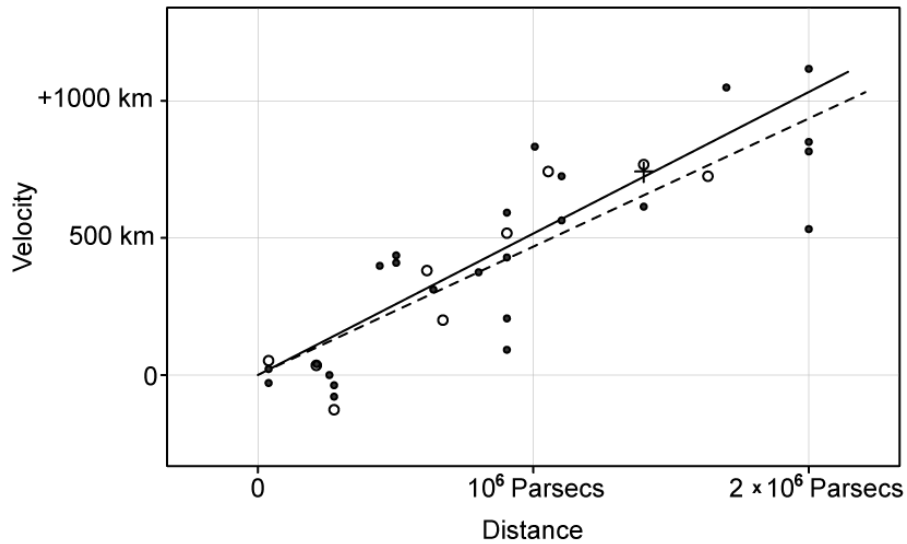


Fig. 1.2

(i)

Using data from Fig. 1.2, calculate and compare Hubble's original estimate for the age of the universe, in years, with our modern estimation.

$$1 \text{ parsec} = 3.09 \times 10^{16} \text{ m}$$

[3]

(ii)

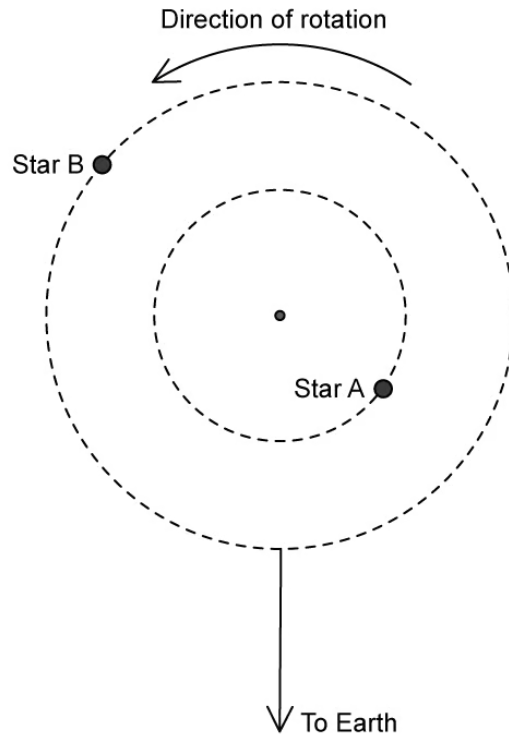
Discuss why, even today, different measurements of the Hubble constant do not agree with each other.

[2]

[5 marks]

Question 2a

Two stars, A and B, in a binary system move in an anti-clockwise direction around a common centre of mass, as shown in Fig. 1.1.

**Fig. 1.1**

On Fig. 1.2a and Fig. 1.2b below, mark the positions of stars A and B corresponding to their spectra as observed on Earth.

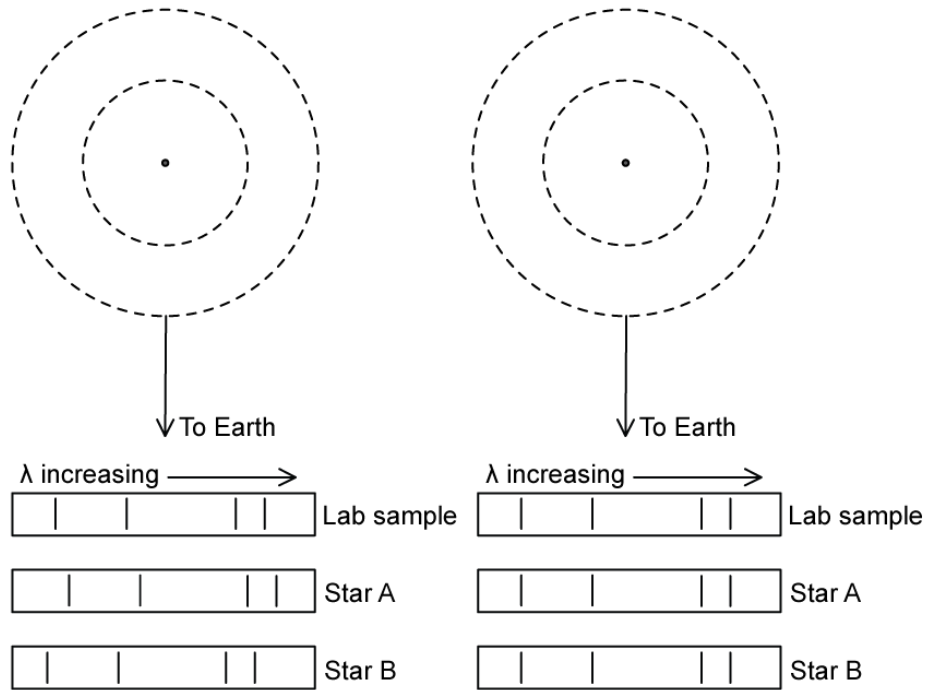


Fig. 1.2a

Fig. 1.2b

[4 marks]

Question 2b

When measured in the laboratory, one of the spectral lines of hydrogen has a wavelength of $4.861 \times 10^{-7} \text{ m}$.

The same hydrogen line in the spectrum of one of the stars fluctuates from its laboratory wavelength by $\pm 0.420 \times 10^{-7} \text{ m}$ and the line in the spectrum of the other star fluctuates by $\pm 0.865 \times 10^{-7} \text{ m}$.

Calculate the velocity of each star in the binary system and identify the faster star and the slower star.

[4 marks]

Question 2c

The orbital period of the binary system about the common centre of mass is 21 days.

Show that the orbital radius of star B is about twice the orbital radius of star A.

[3 marks]

Question 3a

Newton assumed that the universe was infinite, uniform and static. The Big Bang model suggests space and time originated at one point around 14 billion years ago. At this time the temperature was very high.

In 1965, Penzias and Wilson discovered cosmic microwave background (CMB) radiation with a wavelength of around 1 mm.

Determine the current temperature of the CMB.

Surface temperature of the Sun = 5800 K

Peak wavelength emission of the Sun = 500 nm

[2 marks]

Question 3b

Fig. 1.1 shows the map of microwave background radiation from the COBE satellite. It shows the variation in the mean temperature of the CMB in different parts of the sky.

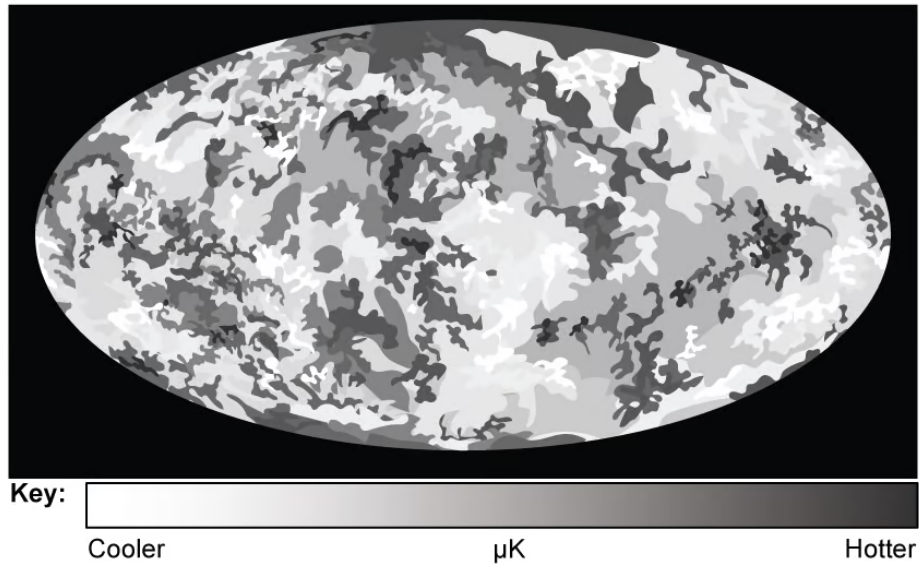


Fig. 1.1

This radiation was first emitted around 300 000 years after the Big Bang, during a stage of the universe's history called the 'era of recombination'.

At this stage, the radiation was in the visible region of the spectrum and had a temperature of around 3000 K.

Determine the factor by which the universe has expanded since the universe was at a temperature of 3000 K.

[3 marks]

Question 3c

Using Fig. 1.1, your answers to **(a)** and **(b)**, and your knowledge of redshift and cosmology

(i)

Describe how the CMB provides evidence for the Hot Big Bang model of the universe.

[2]

(ii)

State and explain what the COBE data in Fig. 1.1 suggests about the temperature variation in the early universe.

[2]

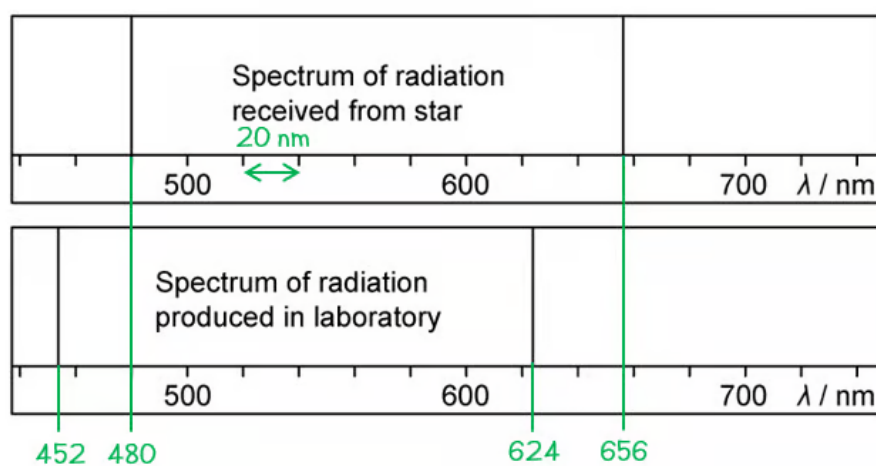
[4 marks]

Model Answers: Hard

1a

(a) Determine the distance of the galaxy from Earth:

Use the line spectra to identify the wavelengths of the emitted radiation:



- (From diagram) $\lambda_{\text{star}} = 480 \text{ nm}$ **AND** $\lambda_{\text{lab}} = 454 \text{ nm}$ [1 mark]
- Shift in wavelength: $\Delta\lambda = 480 - 454 = 26 \text{ nm}$ [1 mark]

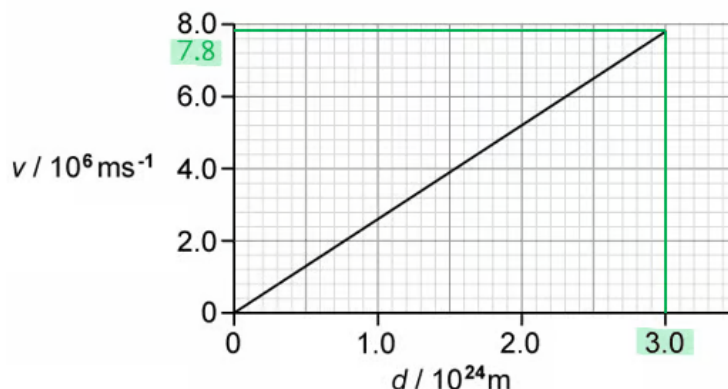
Acceptable range for $\lambda_{\text{star}} = 479 \text{ to } 480 \text{ nm}$

Acceptable range for $\lambda_{\text{lab}} = 452 \text{ to } 456 \text{ nm}$

Determine the recessional velocity of the galaxy:

- Doppler redshift: $\frac{\Delta\lambda}{\lambda} \approx \frac{v}{c} \Rightarrow v = c \times \frac{\Delta\lambda}{\lambda}$
- $v = (3 \times 10^8) \times \frac{26}{454} = 1.72 \times 10^7 \text{ m s}^{-1}$ [1 mark]

Determine the gradient (Hubble constant) from the graph



- Hubble's law: $v = H_0 d$
- $H_0 = \frac{\Delta v}{\Delta d} = \frac{7.8 \times 10^6}{3.0 \times 10^{24}} = 2.6 \times 10^{-18} \text{ s}^{-1}$ [1 mark]

Determine the distance to the galaxy:

- $d = \frac{v}{H_0} = \frac{1.72 \times 10^7}{2.6 \times 10^{-18}} = 6.6 \times 10^{24} \text{ m}$ [1 mark]

[Total: 5 marks]

Make sure to use a **ruler** to identify the wavelengths between those identified on the axis as **accurately** as possible

You could use the **other wavelengths** in your calculation to get the **same answer**:

- $\lambda_{star} = 656 \text{ nm}$:
- Acceptable range: 654 to 658 nm
- $\lambda_{lab} = 624 \text{ nm}$
- Acceptable range: 622 to 626 nm

In the Doppler shift equation $\lambda = \lambda_{lab}$ because this is the **source radiation**, the radiation from the star is compared with this

1b

(b)

(i) Determine the age of the universe when the light from the distant galaxy was

emitted:

List the known quantities from **(a)**:

- Emitted wavelength, $\lambda = 454 \text{ nm}$
- Change in wavelength, $\Delta\lambda = 26 \text{ nm}$
- Hubble constant, $H_0 = 2.6 \times 10^{-18} \text{ s}^{-1}$

Calculate the value of redshift for the galaxy:

- Redshift: $z \approx \frac{\Delta\lambda}{\lambda} = \frac{26}{454} = 0.057$ [1 mark]

Use the expression to determine the expansion rate of the universe:

- Expansion rate: $\frac{R}{R_0} = 1 + z = 1 + 0.057 = 1.057$
- Expansion rate: $\frac{R}{R_0} = \frac{t}{t_0} = 1.057$ [1 mark]

Determine the age of the universe:

- Age of the universe (now): $t = \frac{1}{H_0}$
- $t = \frac{1}{2.6 \times 10^{-18}} = 3.85 \times 10^{17} \text{ s}$
- $t = \frac{3.85 \times 10^{17}}{60 \times 60 \times 24 \times 365} = 12.2 \times 10^9 \text{ years}$ [1 mark]

Determine the age of the universe when the light was emitted:

- Age of the universe (then): $\frac{t}{t_0} = 1.057 \Rightarrow t_0 = \frac{t}{1.057}$
- $t_0 = \frac{12.2 \times 10^9}{1.057} = 11.5 \times 10^9 \text{ years}$ [1 mark]

- Therefore, the universe was 11.5 billion years old when the light was emitted

(ii) The assumption made in the calculation is:

- The rate of expansion (of the universe) is uniform / constant / non-accelerated
OR H_0 is constant over time; [1 mark]

[Total: 5 marks]

The alternative method you could use is to consider how far the light has travelled in light years and subtract this from the age of the universe, this is because

- 1 light year = the **distance** travelled by light in one year
- 1 light year = speed of light (m s^{-1}) $\times$ time (seconds in 1 year)
- 1 light year = $(3 \times 10^8) \times (60 \times 60 \times 24 \times 365)$

Although you are not required to know anything about light years for your exam, using your knowledge of light in this way can help to further your understanding of the topic of cosmology and the vast distances involved

Over a distance of 6.6×10^{24} m (from **(a)**), light has travelled

- distance (ly) =
$$\frac{6.6 \times 10^{24}}{(3 \times 10^8) \times (60 \times 60 \times 24 \times 365)} = 6.98 \times 10^8 \text{ ly}$$

This means the light was emitted **0.7 billion years** ago

At this time, the universe was $(12.2 - 0.7) = \mathbf{11.5 \text{ billion years old}}$

1c

(c)

(i) Determine Hubble's original estimate for the age of the universe:

List the known quantities:

- Age of the universe, $t = 12.2 \times 10^9$ years (from **(b)**)

Determine the gradient of the graph:

- distance (ly) = $\frac{6.98 \times 10^8}{(3 \times 10^8) \times (60 \times 60 \times 24 \times 365)} = 6.98 \times 10^8 \text{ ly}$

This means the light was emitted **0.7 billion years** ago

At this time, the universe was $(12.2 - 0.7) = \mathbf{11.5 \text{ billion years old}}$

1c

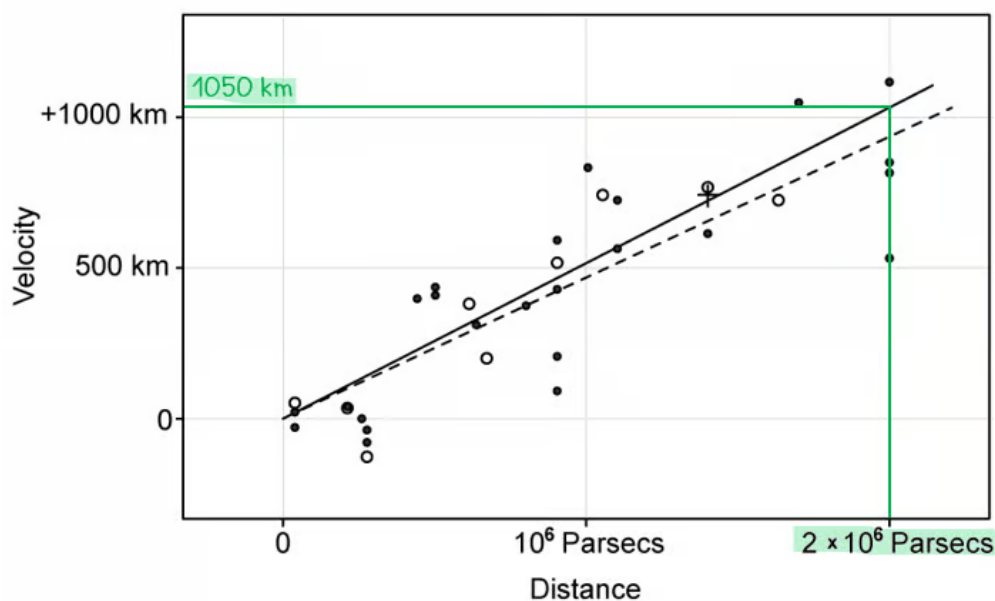
(c)

(i) Determine Hubble's original estimate for the age of the universe:

List the known quantities:

- Age of the universe, $t = 12.2 \times 10^9$ years (from **(b)**)

Determine the gradient of the graph:



- $H_0 = \frac{\Delta v}{\Delta d} = \frac{1050 \times 10^3}{(2 \times 10^6) \times (3.09 \times 10^{16})}$

- $H_0 = 1.699 \times 10^{-17} \text{ s}^{-1}$ [1 mark]

Determine Hubble's estimate for the age:

- $t = \frac{1}{H_0} = \frac{1}{1.699 \times 10^{-17}} = 5.89 \times 10^{16} \text{ s}$
- $t = \frac{5.89 \times 10^{16}}{60 \times 60 \times 24 \times 365} = 1.9 \times 10^9 \text{ years}$ [1 mark]

Acceptable range for t (if upper line used) = 1.8 to 2.0 billion years

Acceptable range for t (if lower line used) = 2.1 to 2.3 billion years

Compare to the modern estimate:

- Hubble's original estimate is about $(12.2 - 1.9 =) 10$ billion years lower than our modern estimate of the universe's age; [1 mark]

(ii) Measurements of the Hubble constant do not agree with each other because:

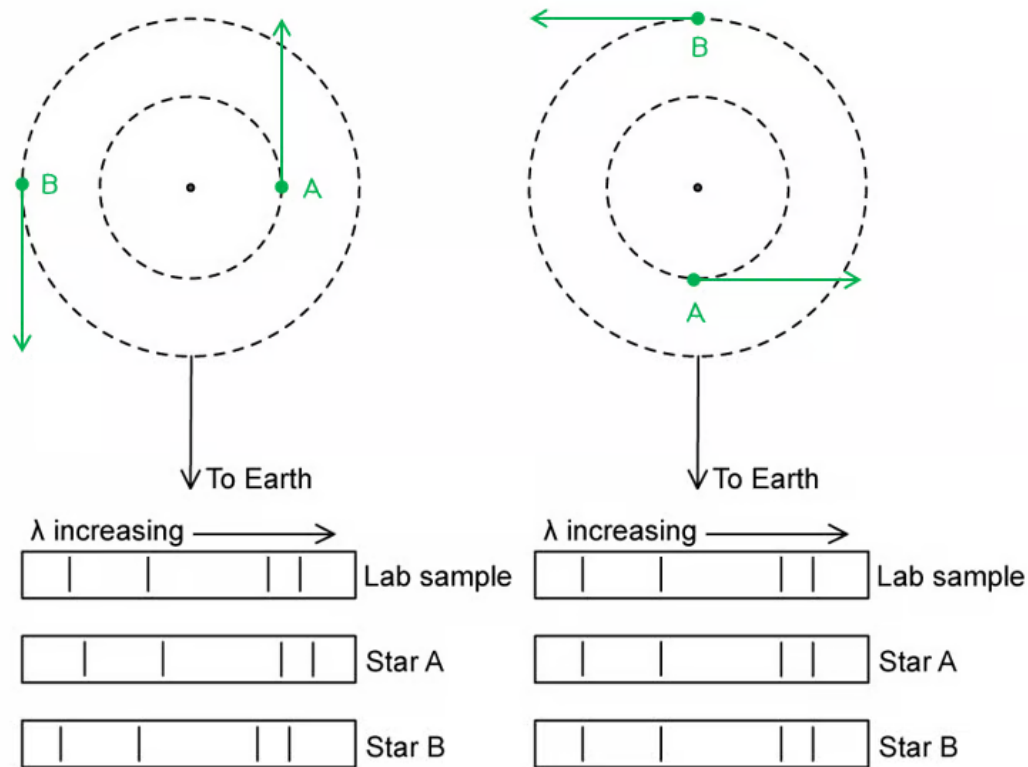
Any **two** from:

- Measurements from distant galaxies have large uncertainties; [1 mark]
- Galaxies that are close by (enough to observe finer detail) may have peculiar motion e.g. galaxies moving towards us (blueshifted) / (two or more) galaxies orbiting around a centre of mass etc; [1 mark]
- Galaxies were not formed at the beginning of the universe, hence H_0 is a lower limit; [1 mark]
- The rate of expansion may not be constant / may be accelerating; [1 mark]

[Total: 5 marks]

2a

(a) The position of the stars corresponding to their spectra as observed on Earth is as follows:



On Fig. 1.2a...

- Star A (inner orbit) is marked to the right of the centre of mass with an arrow directed away from the Earth; [1 mark]
- Star B (outer orbit) is marked to the left of the centre of mass with an arrow directed towards the Earth; [1 mark]

On Fig. 1.2b...

- Both stars are marked directly above/below the centre of mass (in their respective orbits); [1 mark]
- Both arrows are perpendicular relative to the Earth in an anti-clockwise direction; [1 mark]

[Total: 4 marks]

2b

(b) Calculate the velocity of the each star:

List the known quantities:

- Wavelength of spectral line in a lab, $\lambda = 4.861 \times 10^{-7} \text{ m}$

- Change in wavelength of slower star, $\Delta\lambda = 0.420 \times 10^{-7} \text{ m}$
- Change in wavelength of faster star, $\Delta\lambda = 0.865 \times 10^{-7} \text{ m}$
- Speed of light, $c = 3.0 \times 10^8 \text{ m s}^{-1}$

Calculate the velocity of the slower star:

- Doppler redshift: $\frac{\Delta\lambda}{\lambda} \approx \frac{v}{c} \Rightarrow v = c \times \frac{\Delta\lambda}{\lambda}$
- Since $\Delta\lambda \propto v$, for the slower star: $\Delta\lambda = 0.420 \times 10^{-7} \text{ m}$
- $v = (3.0 \times 10^8) \times \frac{0.420}{4.861}$ [1 mark]
- $v = 2.59 \times 10^7 \text{ m s}^{-1}$ [1 mark]

Calculate the velocity of the faster star:

- For the faster star: $\Delta\lambda = 0.865 \times 10^{-7} \text{ m}$
- $v = (3.0 \times 10^8) \times \frac{0.865}{4.861}$
- $v = 5.34 \times 10^7 \text{ m s}^{-1}$ [1 mark]

Identify the velocities of stars A and B:

- Orbital speed: $v = \frac{2\pi r}{T} \Rightarrow v \propto r$
- Star A has the smaller orbital r , hence it is the slower star / $v_A = 2.59 \times 10^7 \text{ m s}^{-1}$
OR
- Star B has the larger orbital r , hence it is the faster star / $v_B = 5.34 \times 10^7 \text{ m s}^{-1}$; [1 mark]

[Total: 4 marks]

2c

(c) Show that the orbital radius of star B is about twice the orbital radius of star A:

List the known values:

- Period of system, $T = 21 \text{ days} = 21 \times (60 \times 60 \times 24) = 1.81 \times 10^6 \text{ s}$
- Velocity of star A, $v_A = 2.59 \times 10^7 \text{ m s}^{-1}$ (from **(b)**)
- Velocity of star B, $v_B = 5.34 \times 10^7 \text{ m s}^{-1}$ (from **(b)**)

Determine the orbital radius of star A:

- Orbital speed: $v = \frac{2\pi r}{T} \Rightarrow r = \frac{vT}{2\pi}$
- $r_A = \frac{(2.59 \times 10^7)(1.81 \times 10^6)}{2\pi} = 7.46 \times 10^{12} \text{ m}$ [1 mark]

Determine the orbital radius of star B:

- $r_B = \frac{(5.34 \times 10^7)(1.81 \times 10^6)}{2\pi} = 15.4 \times 10^{12} \text{ m}$ [1 mark]

Determine the ratio of the orbits:

- Ratio: $\frac{r_B}{r_A} = \frac{15.4 \times 10^{12}}{7.46 \times 10^{12}} = 2.064 = 2.1$ [1 mark]

[Total: 3 marks]

3a

(a) Determine the current temperature of the CMB:

List the known quantities:

- Surface temperature of the Sun, $T_{Sun} = 5800 \text{ K}$
- Peak wavelength emission of the Sun, $\lambda_{Sun} = 500 \text{ nm}$
- Peak wavelength emission of the CMB, $\lambda_{CMB} = 1 \text{ mm}$

Use Wien's law to determine the temperature of the CMB:

- Wien's law: $T \propto \frac{1}{\lambda_{max}}$
- Therefore, $\frac{T_{CMB}}{T_{Sun}} = \frac{\lambda_{Sun}}{\lambda_{CMB}} \Rightarrow T_{CMB} = T_{Sun} \times \frac{\lambda_{Sun}}{\lambda_{CMB}}$
- $T_{CMB} = 5800 \times \frac{500 \times 10^{-9}}{1 \times 10^{-3}}$ [1 mark]
- $T_{CMB} = 2.9 \text{ K}$ [1 mark]

[Total: 2 marks]

3b

(b) Determine the factor by which the universe has expanded:

List the known quantities:

- Initial temperature, $T_{then} = 3000 \text{ K}$
- Final temperature, $T_{now} = 2.9 \text{ K}$ (from **(a)**)

Use the relationship between energy and temperature for a photon:

- Energy-temperature relation for photons: $E = kT$
- $\frac{\text{energy of photons at 3 K}}{\text{energy of photons at 3000 K}} = \frac{kT_{now}}{kT_{then}} = \frac{2.9}{3000}$ [1 mark]
- $\frac{\text{energy of photons at 3 K}}{\text{energy of photons at 3000 K}} = 0.00097$ [1 mark]

Use the equation for photon energy and de Broglie wavelength:

- Photon energy: $E = \frac{hc}{\lambda} \Rightarrow \lambda \propto \frac{1}{E}$

$$\bullet \frac{\lambda \text{ of photons at } 3 \text{ K}}{\lambda \text{ of photons at } 3000 \text{ K}} = \frac{1}{0.00097} = 1030 \text{ [1 mark]}$$

[Total: 3 marks]

Remember, redshift tells us the **factor of expansion** of the universe at different points in time/space, so in a question like this, you need to think about the

relationships you know to try and get a ratio in terms of $\frac{\Delta\lambda}{\lambda}$

Since the difference in wavelengths between these temperatures is so large,
 $\Delta\lambda \approx \lambda \text{ of photons at } 3 \text{ K}$

3c

- (c)
- (i) The CMB provides evidence for the Hot Big Bang model of the universe because:

EITHER

- The wavelength of the CMB has been redshifted by an expanding and cooling universe; [1 mark]
- The shorter wavelength in the past indicates the temperature must have been very high at the beginning; [1 mark]

OR

- The temperature of the CMB has cooled considerably from $t = 300\,000$ to now and even more since the Big Bang; [1 mark]
- This cooling was caused by the expansion of the universe / the stretching of spacetime; [1 mark]

- (ii) The COBE data suggests that:

- The wavelength / intensity of the CMB is (almost) the same in all directions **OR** there is very little temperature variation in the CMB (on the scale of μK); [1 mark]
- This indicates that there was very little temperature variation in the early universe too; [1 mark]

